REPUBLIC OF CAMEROON
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HIGHER INSTITUTE OF PROFESSIONAL EXCELLENCE (HIPE)

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DESIGN AND IMPLEMENTATION OF AN ONLINE VOTING SYSTEM

CASE STUDY: HIGHER INSTITUTE OF PROFESSIONAL EXCELLENCE (HIPE)

AN INTERNSHIP REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE AWARD OF THE HIGHER NATIONAL DIPLOMA (HND)

IN SOFTWARE ENGINEERING

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ACADEMIC SUPERVISOR

MR DOGO VALERY

ACADEMIC YEAR 2025/2026

DESIGN AND IMPLEMENTATION OF AN ONLINE VOTING SYSTEM.

CERTIFICATIONS

This is to certify that this internship report entitled "Design and Implementation of an Online Voting System" was originally written by EYONG NICOLINE ATEM to meet the partial requirements for the award of the Higher National Diploma (HND) in Software Engineering and is hereby approved for assessment of the requirements for the award of the HND in Software Engineering.

ACADEMIC SUPERVISOR

MR DOGO VALERY

DATE: ______________________

SIGNATURE: _________________

PROFESSIONAL SUPERVISOR

MR TANTE HENRY

DATE: ______________________

SIGNATURE: _________________

DEDICATION

To the Eyong and Eshuali Family

ACKNOWLEDGEMENT

This piece of work has been difficult and filled with many obstacles but by the grace of God, I was able to accomplish the task. I give thanks to God Almighty who guided me throughout this project.

- To the Director of the HIGHER INSTITUTE OF PROFESSIONAL EXCELLENCE, for having given us the means to carry out our project.
- To Mr. DOGO, I would also like to express my gratitude for your expert advice, collaboration and unwavering support.
- To Mr. TANTE HENRY, I would like to express my gratitude for your expert advice, and collaboration.
- I would like to thank my mom Mrs. AKO GLADYS for her unconditional support, patience and encouragement.
- I would like to thank my family and friends for their unconditional support, patience and encouragement.
- To all my teachers and all the staff of the HIGHER INSTITUTE OF PROFESSIONAL EXCELLENCE
- TO MY STUDY GROUP for their great support, patience and collaboration.

ABSTRACT

The Online Voting System is a web-based platform designed to make elections more transparent, secure, and efficient. Traditional voting methods often face problems such as ballot loss, manipulation, long queues, and delayed results. This project addresses these issues by allowing registered students of HIPE to vote electronically from any location using secure login credentials.

Developed with HTML and CSS, **bcryptjs, cloudinary, cors, dotenv, express, express-fileupload, jsonwebtoken, mongoose,** and **uuid**, the system provides a professional, responsive, and user-friendly interface accessible from any internet-enabled device. It ensures data confidentiality, voter authentication, and accurate result computation. By automating the voting and counting process, the system supports the HIPE Electoral Commission in conducting free and fair elections while enhancing student's and participation in the election process.

RÉSUMÉ

Le **Système de Vote en Ligne** est une plateforme web conçue pour rendre les élections plus transparentes, sécurisées et efficaces. Les méthodes de vote traditionnelles rencontrent souvent des problèmes tels que la perte des bulletins, la manipulation, les longues files d'attente et le retard dans la publication des résultats. Ce projet répond à ces difficultés en permettant aux étudiants inscrits de **HIPE** de voter électroniquement depuis n'importe quel endroit à l'aide d'identifiants de connexion sécurisés.

Développé avec **HTML et CSS**, **bcryptjs**, **cloudinary**, **cors**, **dotenv**, **express**, **express-fileupload**, **jsonwebtoken**, **mongoose** et **uuid**, le système offre une interface professionnelle, réactive et conviviale, accessible depuis tout appareil connecté à Internet. Il garantit la confidentialité des données, l'authentification des électeurs et un calcul précis des résultats. En automatisant le processus de vote et de dépouillement, le système soutient la Commission Électorale de HIPE dans l'organisation d'élections libres et équitables, tout en renforçant la participation et l'engagement des étudiants dans le processus électoral.

PREFACE

Higher Institute of Professional Excellence (HIPE) was created on the 22 May 2017 with authorization number No 17-0521/L/MINESUP/DDES/ESUP/SDA/OPC to run the following specialties; was created to permit young Cameroonians to acquire the appropriate skills needed by employers in the competitive job market. HIPE is located at Ancient Route Bonabéri, adjacent to pharmacie de Bonambappe_Douala the quest for professionalism of young graduates in the job market is the main reason that led to the creation of Higher Institute of Professional Excellence (HIPE).

SCHOOL OF BUSINESS	SCHOOL OF ENGINEERING	SCHOOL OF MEDICAL SCIENCE	SCHOOL OF TOURISM

➢ Accountancy ➢ Banking and Finance ➢ Marketing and Management ➢ Logistics and Transport ➢ Human Resource Management ➢ Executive ➢ Secretarial Management ➢ Project Management	➢ Civil Engineering ➢ Computer ➢ Engineering ➢ Electrical Engineering ➢ Mechanical Engineering ➢ Word Technology	➢ Laboratory ➢ Technology ➢ Nursing ➢ Midwifery ➢ Physiotherapy	➢ Hotel ➢ management ➢ Tourism ➢ Bakery and Food Processing ➢ Fashion, Textile ➢ Beauty Care ➢ Cosmetics

Table 1: School of study

IDENTIFICATION

SECTION	DETAILS
COMPANY NAME	TTSET GLOBAL LTD
DATE OF CREATION	YEAR 2021
COUNTRY OF REGISTRATION	CAMEROON
LEGAL STRUCTURE	LTD
INDUSTRY SECTOR	IT
HEAD OFFICE ADDRESS	DOUALA-BONABERI
CITY/STATE	LITTORAL
COUNTRY	CAMEROON
TELEPHONE NUMBER	676080714/679303247
EMAIL ADDRESS	ttsetglobal@gmail.com
WEBSITE	https://www.ttsetglobal.com
CAPITAL	10,000,000 XAF
NUMBER OF EMPLOYEES	10
PRODUCT/SERVICES OFFERED	IT SERVICES

Table 2: Identification form

LIST OF ABBREVIATIONS

IT: Information Technology.

CEO: Chief Executive Officer.

COO: Chief Operating Officer.

CTO: Chief Technology Officer.

SEO: Search Engine Optimization.

HTML: HyperText Markup Language.

CSS: Cascading Style Sheets.

DoS: Denial-of-Service.

DNS: Domain Name System.

WIFI: Wireless Fidelity.

VLAN: Virtual Local Area Network.

DHCP: Dynamic Host Configuration Protocol.

OSPF: Open Shortest Path First.

IP: Internet Protocol.

MySQL: A combination of “My”, the name of co-founder Michael Widenius’s daughter, and “SQL”, abbreviation for Structured Query Language.

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GENERAL INTRODUCTION

Background and Context of the Study

Elections are a cornerstone of democracy, allowing citizens to choose their leaders and influence policy decisions. Traditionally, voting has been conducted using paper ballots at polling stations. While this method has been effective for decades, it is associated with several challenges, including long queues, human errors, ballot misplacement, fraud, and delays in result declaration. These problems reduce public confidence in the electoral process and sometimes discourage citizen participation.

With the rise of information and communication technologies, online voting systems have emerged as a practical solution to these challenges. An Online Voting System allows registered citizens to cast their votes electronically from any location, ensuring convenience, accessibility, and transparency. This system not only addresses the inefficiencies of manual voting but also aligns with global trends toward digital transformation in public services, reflecting a move toward more efficient and citizen-friendly.

Statement of the Problem

Despite the importance of elections, traditional paper-based systems continue to face inefficiencies and risks. Physical ballots can be misplaced, tampered with, or counted incorrectly, while long queues and manual procedures discourage voter participation. These problems compromise transparency, accuracy, and public trust in election outcomes. There is therefore an urgent need for a secure, reliable, and accessible system. An Online Voting System provides a solution by automating the casting, recording, and counting of votes, reducing errors and delays, and enhancing the credibility of the electoral process.

Research Questions

1. **Main question:** How can an online Voting System improve transparency, security, and efficiency in national elections?
2. **Subsidiary questions:**
    - ❖ How can citizens vote remotely without compromising election integrity?
    - ❖ How can accurate vote tallying and result declaration be ensured electronically?

Objectives of the Study

Main Objective: To develop a secure, reliable, and user-friendly Online Voting System that improves electoral processes.

Subsidiary Objectives:

1. Provide a platform that allows registered citizens to vote electronically from any location.
2. Ensure accurate vote recording, automated counting, and transparent result publication.

By achieving these objectives, the system aims to reduce inefficiencies, improve accessibility, and enhance public trust in elections.

Methodology

The methodology used in this study includes **research, system design, implementation, and testing.**

- **Research:** Reviewed literature on traditional and electronic voting systems, as well as case studies of electoral processes in Higher Institute of Professional Excellence (HIPE).
- **System Design:** Defined requirements, created data flow and use case diagrams, and designed wireframes for the user interface.
- **Implementation:** Developed a web-based application using **HTML** and **CSS**, providing a responsive, professional, and user-friendly interface. Backend integration with technologies like **bcryptjs, cloudinary, cors, dotenv, express, express-fileupload, jsonwebtoken, mongoose,** and **uuid** will be added in the next development phase.

- **Testing:** Conducted usability and functionality tests to ensure the system operates smoothly and meets user expectations.
- **Result Presentation:** Demonstrated system features through interface screenshots, vote submissions, and sample reports of voting results.

Structure of the Work

The report is organized into four chapters to provide a comprehensive understanding of the project:

PART ONE: THE INTERNSHIP FRAMEWORK

presents the company hosting the internship, including its creation, mission, structure, products, services, and operations. It also describes the student's internship activities, weekly tasks, observations, skills acquired, and the justification for choosing the project topic.

CHAPTER ONE: GENERAL PRESENTATION OF THE COMPANY

This chapter is divided into two sections. Section one talks about the creation and historical evolution of TTSET GLOBAL while section two talks about Organization and operation of the company.

I.1. CREATION AND HISTORICAL EVOLUTION OF THE COMPANY

I.1.1. History and creation of the company:

TTSET GLOBAL was founded in 2021 by two partners who envisioned creating a leading IT service in Cameroon. This was intended to offer comprehensive IT solutions to businesses across various industries. The departments/Team's name was established years ago to focus on specific area. The company easy gained traction by focusing on core services such as managed IT services, cybersecurity and cloud service. This rapid adoption by clients demonstrated the immediate need for reliable IT solutions in the region, establishing TTSET Global as a trusted partner.

I.1.2. Evolution:

Furthermore, recognizing the evolving market, TTSET GLOBAL expanded its services portfolio to include software development, IT consulting, data analysis services. This phase also saw the company introducing digital marketing, and CEO services which position them as one- stop shop for businesses seeking digital transformation. In 2023 TTSET Global further decided to diversify and meet client demands as to venture into photography and videography services. This expansion hereby allows the company offer a unique of IT and Multimedia services, making it a leading provider of digital content solutions alongside with core IT services.

The internship program was initiated years ago to provide students with hands-on experience in the field.

I.3. The products and services of the company

TTSET Global is aimed at to create a management system for schools, online marketing and graphical designs used in creating automated systems for programming, digital marketing and SCO services, hardware and software, procurements, cloud services, cyber security services remote work solutions and photograph and videography services. It also develops websites of all types, which is one of its numerous services where they train students and people in order to become a good web designer and web developer in future. Though not fully established, the company also obtain a digital expansion and advancement towards full growth where the company seeks to acquire its own channel (TTSET GLOBAL TV).

I.2. ORGANIZATION AND OPERATION OF THE COMPANY

TTSET Global operates with a well-defined organization structure led by three co-founders who serve as the top management team. Each co-founder oversees a specific aspect of the business to ensure seamless operations, service delivery, and strategic growth.

I.2.1. The Organizational Structure of the company

TTSET Global follows a well-structured managerial system which starts from the top management, departments and reporting structure and operational

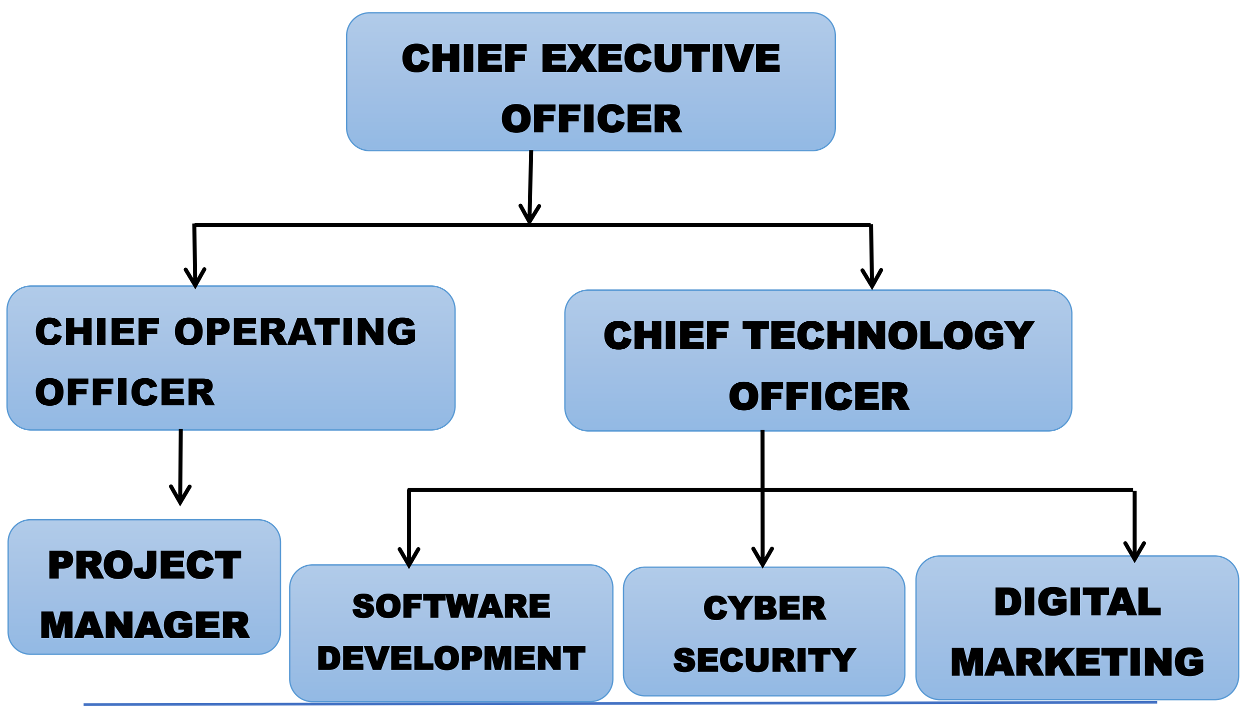


Figure 1: Organizational chart

1. Top management

- Founder A-chief executive officer (CEO): Oversees the overall vision, strategy, and business development. Responsible for driving the company's growth and exploring new business opportunities.
- Founder B-chief operating officer (COO): Manages day-to-day operations, ensures efficient service delivery, oversees project management, and coordinates inter-departmental activities.
- Founder C-chief technology officer (CTO): Leads technology strategy, software development, Cybersecurity initiatives, and innovation within the company. Departments and reporting structure, the company's departments are structured under the leadership of each co-founder, ensuring clear lines of responsibility and effective communication.

1.1. Department under the CEO's supervision:

- **Sales and marketing department:** Responsible for generating leads, conducting marketing research, managing digital marketing campaigns, and developing sales strategies.

- **Client relation and account management:** Manages existing clients, maintains client satisfaction, gathers feedback and ensures ongoing business relationship.

1.2. Department under the COO's supervision:

- **IT service delivery department:** Handles all managed IT services, including cybersecurity, cloud services, and infrastructure management. Ensures smooth implementation and monitoring of client IT systems.
- **Photography and videography department:** Provides multimedia content creation, including photography, video production, and editing for clients.
- **Training and support department:** Offers IT training sessions, workshops and ongoing technical support to clients.

1.3. Department under the CTO's supervision:

- **Software development department:** Develops custom software, mobile applications, and handles software maintenance projects.
- **Cybersecurity department:** Implements security policies, monitors threats, and conducts vulnerability assessment to protect clients' data and infrastructure.

- ❖ **Data analysis department:** Manages data analytics projects, processes data, and delivers actionable insight for client businesses. Operational procedures.
- ❖ **Overview of workflows and departmental functions:** TTSET GLOBAL has developed efficient Workflows to ensure smooth operations and high-quality services.
- ❖ **Service delivery:** Each department follows well-defined procedures for optimal efficiency:

❖ **Sales and Marketing Department:**

- ❖ **Lead generation:** Identify potential clients through market research and digital marketing campaigns.
- ❖ **Sales funnel management:** Nurtures leads through the sales process, converting prospects into clients.

❖ **Client relations and account management:**

- ❖ **Client on-boarding:** Ensures smooth on-boarding for new clients, introducing them to TTSET GLOBAL's range of services.
- ❖ **Account management:** Regularly engages with clients to ensure service satisfaction, identify additional needs, and offer tailored solutions.

❖ **IT service delivery department:**

- ❖ **Project planning:** Develops project plans with time lines, resources, and milestones for IT service implementation.
- ❖ **Service delivery:** Execute projects, monitors systems, and provides ongoing maintenance to meet client requirements.

- ❖ **Troubleshooting and support:** Provides quick resolution for any technical issues faced by clients, ensuring minimal downtime.

❖ **Photography and Videography department:**

- ❖ **Content planning:** Works with clients to understand their multimedia needs and develop content plan.
- ❖ **Production and Editing:** Conducts photo shoots and video recordings, followed by editing and post production to deliver high-quality multimedia content.

❖ **Training and Support Department:**

- ❖ **Training sessions:** Develops training materials and conducts IT workshops for clients, both in-person and online.
- ❖ **Support services:** Provides 24/7 technical support to clients, addressing any IT-related issues they encounter.

❖ **Software Development Department:**

- ❖ **Requirement Analysis:** collaborates with clients to understand software needs and develop project specifications.
- ❖ **Development Circle:** Follow Agile methodologies, delivering software projects in phases and ensuring continuous client feedback and iteration.
- ❖ **Testing and Deployment:** conduct thorough testing before deploying software solutions, ensuring quality and functionality.

❖ **Cybersecurity Department:**

- ❖ **Security Assessment:** conduct risk assessments and vulnerability testing for clients' IT systems.

    - **Implementation of Security Solutions:** Deploys security measures such as firewalls, encryption and intrusion detection systems.
    - **Ongoing Monitoring:** Monitors client system for potential threats and provides timely responses to security incidents.
- **Analysis and Reporting:** Uses advanced analytics tools to generate insights and create reports for client, aiding in decision-making. The collaborative nature of the TTSET GLOBAL's departments ensures that clients receive comprehensive IT solutions tailored to their needs. The company's efficient operational procedures allow for seamless delivery of services, maintaining its reputation as a trusted IT partner in Cameroon, Littoral Douala.

CHAPTER TWO: THE EVOLUTION OF INTERNSHIP ACTIVITIES

INTRODUCTION:

This chapter presents the PRESENTATION OF THE SERVICE at TTSET GLOBAL and the EVOLUTION OF THE INTERNSHIP ACTIVITIES AND JUSTIFICATION OF TOPIC.

II.1. PRESENTATION OF THE SERVICE HOSTING AT THE COMPANY

This session explains how we were received in the company by the major departments.

II.1.1. The Receptionist

On the first day of the internship, we were warmly received by the receptionist who explained the functions of the different departments and later took us to the director.

II.1.2. The Software Development Department

We were received by the head of Software Engineering Development we got to know all the company staff and meet faces for the first time. During this time, we saw already developed applications done by former interns. We discussed with our various superiors and they talked about the different tasks which will be carried on our internship placement. To add, we were asked questions on our skills gotten from our various schools and was given an opportunity to showcase it to them during our stay there.

II.3. The Department Network and Security

We had the opportunity to attend the Network and Security sessions where the Head of Network and Security project explained the need for security in personal and organizational networks. He explained about malware, phishing, and man-in-the-middle attacks, the Denial-of-service (DoS) attacks, insider attack, Domain Name System (DNS) spoofing.

II.1.4. Internal Environment of the Company

TTSET GLOBAL operates by the following means.

- ❖ Punctuality and reliability of staff members.
- ❖ It was more of practical than theoretical.
- ❖ TTSET GLOBAL staff constantly update us in the various programming languages to be used while coding which was Cascading Style Sheet, JavaScript, Hypertext preprocessor, Python.
- ❖ Also, we were supplied with constant connection services with the use of Wireless Fidelity (WIFI).

❖ The Supplier

They supply TTSET with requirements such as computer consumables and maintain professional relationships.

❖ Customer

They are classified into two categories:

- **Regular:** They are registered in a solid company database. These are clients with whom the company constantly deals with (clients with contracts).
- **Irregular:** They request occasional 4e services without any contractual agreement.

II.2. EVOLUTION OF THE INTERNSHIP ACTIVITIES AND JUSTIFICATION OF TOPIC

I.2.1. Task Carried Out

Summaries and records of activities were taken on a weekly basis as shown below.

Week	Activity	Lesson Learned
From 7th July to the 11th of July.	We had our first contact with our working environment and staffs which were to train us for stay in the establishment.	We learned on what all our working enterprise activities focuses on.
From 14th July to the 18th of July.	We also had our first class on web programming.	We learnt on how to create our portfolio website,

		landing page, adding modal to our portfolio website, creating repository on Github, and deploy our project on Netlify.
From 21st July to the 25th of July.	We also had our first class in Networking, Cybersecurity, and Cloud computing. Also basic Kali Linux commands.	We also learnt types of Malware, Cybersecurity goals, key aspect of networking, classification of cybersecurity tools, types of social engineers, Kali Linux commands or Terminal.
From 28th July to the 1st of August	We also intensified our skills on Networking commands, designed, configured and send packages on a simple network. To add, we created wireless Virtual Local Area Network (VLAN) and Mobile hotspot	We learnt on how to design networks, configured VLANs, Dynamic Host Configuration Protocol (DHCP), Open Shortest Path First routing protocol (OSPF), and to assign the various classes of Internet Protocol address (IP Address). Furthermore, we leant how to create wireless (VLAN) and mobile hotspot connecting to various computers.

From 4th August to the 11th of August.	We also controlled phishing awareness simulation focused on detection and prevention.	We learnt phishing awareness detection and prevention.
From 15th August to the 17th of August.	Further, we create malware using Kali Linux and Metasploitable, we also created sound sensor and fire alarm system using Arduino.	Malware project helped us to spy in someone's device, and also the fire alarm system helped us detect or signal us where there is a fire around the system.
From 25th August to the 29th August	Execution of final internship project.	Errors from our projects were corrected by the panel on the day of the internship defense.

Table 3: Evolution of Internship activities

I.2.3. Justification of Topic

This study is important because the proposed **Online Voting System** will help to enhance transparency, security, and efficiency in the electoral process in the Higher Institute of Professional Excellence (HIPE). It aims to modernize the traditional voting process by introducing a secure, reliable, and accessible digital platform for citizens to cast their votes from any location.

To Citizens (Voters):

- **Accessibility:** Allows participation by people in remote areas or those with mobility challenge.
- **Time saving:** Reduces long queues and delays commonly experienced at polling centers.
- **Transparency:** Voters can verify that their votes are recorded and counted correctly, increasing confidence in the process.

To the Electoral Commission:

- **Efficiency:** Automates vote recording, counting, and result declaration, reducing human error and manual workload.
- **Data security:** Uses encryption and authentication to ensure vote integrity and prevent tampering.
- **Cost reduction:** Minimizes expenses on printing, logistics, and ballot distribution.
- **Faster result publication:** Electronic counting allows real-time announcement of results.

To the Institution:

- **Enhanced credibility:** Builds public trust through transparent and auditable election processes.
- **Digital transformation:** Promotes e-governance and modernization of national systems.
- **Environmental sustainability:** Reduces paper usage and physical waste during elections.

- **Improved participation:** Encourages higher voter turnout due to convenience and accessibility.

To System Developers (Technical Team):

- **Skill development:** Provides experience in web technologies, database security, and network management.
- **Innovation:** Encourages the design of scalable, secure applications suitable for national use.
- **Research and development:** Contributes to the growing field of online voting and cybersecurity.

PARTIAL CONCLUSION

This chapter highlights the justification for developing an Online Voting System and the importance of implementing secure electronic voting nationwide.

During the internship, I acquired valuable skills such as:

- Understanding of **web development** technologies (HTML, CSS, JavaScript, PHP, MySQL.
- Knowledge of **networking and cybersecurity** concepts for secure data transmission.
- Experience in **system design, implementation, and testing** of online applications.

- ❖ Ability to create digital solutions that enhance **transparency, accessibility, and efficiency** in governance.

CHAPTER THREE: LITERATURE REVIEW, MATERIALS AND

III.1. LITERATURE REVIEW

INTRODUCTION

An online voting system enables voters to cast secure electronic votes over the internet, improving efficiency and accessibility. Studies indicate that such systems reduce voting errors, speed up result processing, and increase voter participation. However, security, vote confidentiality, and system integrity remain major challenges. This project builds on prior research to develop a secure and reliable online voting system.

EVOLUTION OF ONLINE VOTING SYSTEM

Voting systems have evolved to improve speed, accuracy, and accessibility. Initially, **manual paper-based voting** was used, which was slow, costly, and prone to errors. Later, **electronic voting machines** were introduced to automate counting and reduce human mistakes, though voters still needed to be physically present. With the rise of the internet, **online voting systems** emerged, allowing secure remote voting and increasing voter participation. Modern systems emphasize **security, authentication, and data integrity** to ensure trust and reliability.

KEY FEATURES AND FUNCTIONALITY

An online voting system is designed to make voting **secure, efficient, and user-friendly**. Its key features and functionalities include:

- **Voter Registration:** Allows eligible voters to register and create accounts securely (Alvarez & Hall, *Electronic Elections*, March 2008).
- **Authentication and Security:** Uses usernames, passwords, or biometric verification to ensure only authorized voters can access the system (Rubin, *Security Considerations for Remote Electronic Voting*, October 2002).
- **Vote Casting:** Enables voters to select candidates or options electronically from any location.
- **Vote Storage and Integrity:** Stores votes securely in a database with encryption to prevent tampering (Kohno et al., *Analysis of Electronic Voting Systems*, July 2004).

- **Result Computation and Display:** Automatically counts votes and displays results accurately and quickly (Trechsel & Vassil, *Internet Voting in Europe*, June 2010).
- **User-Friendly Interface:** Simple and intuitive design for easy navigation by voters of all ages.
- **Audit and Verification:** Provides mechanisms to verify votes and maintain transparency in the election process.

BENEFITS OF AN OLINE VOTING SYSTEM

- **Convenience and Accessibility:** Voters can cast their votes from anywhere at any time, reducing the need to travel to polling stations.
- **Faster Results:** Votes are counted automatically, which significantly reduces the time needed to announce results.
- **Reduced Costs:** Minimizes expenses on printing ballots, hiring polling staff, and transporting materials.
- **Accuracy and Reliability:** Electronic counting reduces human errors associated with manual vote tallying.
- **Increased Participation:** Easier access encourages more voters to participate, improving overall turnout.
- **Security and Transparency:** Proper authentication and encrypted vote storage protect against tampering and ensure trust.

CHALLENGES AND ISSUES

Researchers acknowledge certain challenges and issues in online voting systems, such as:

- **Security risks:** Vulnerability to hacking, unauthorized access, and vote tampering.
- **Authentication problems:** Ensuring only eligible voters can vote and preventing duplicate votes.
- **Data privacy concerns:** Protecting voter information and vote confidentiality.
- **Internet dependency:** Limited access in areas with poor connectivity can reduce participation.
- **Technical trust:** Users may distrust electronic systems compared to traditional voting.
- **System errors:** Bugs or software failures can affect voting and result computation.

CONCLUSION

Online voting systems have evolved significantly, transitioning from traditional paper-based methods to electronic and internet-based platforms. The literature highlights their key features, benefits, and challenges. These systems improve voting efficiency, accessibility, accuracy, and result processing while presenting security, privacy, and trust-related challenges. Future research should focus on emerging technologies such as **artificial intelligence** and **blockchain** to enhance security, transparency, and reliability in online voting systems.

III.1.2. SIMILAR STUDIES

During the development of this project, research was conducted at both **national and international levels** to improve system performance and minimize vulnerabilities. Several projects similar to this system were reviewed, including **online voting platforms**, **e-election systems**, and **secure polling web applications**.

After conducting the study, the following findings were observed:

National Level

Strong Points

- ❖ Creates voter awareness
- ❖ Saves time
- ❖ Reduces operational costs
- ❖ Improves convenience and reduces stress

Weak Points

- ❖ Low level of user interaction in some desktop-based systems
- ❖ Limited server capacity and infrastructure in some African regions

- Risk of data loss due to poor backup systems

III.2. MATERIALS AND METHODS

II.2.1. REQUIREMENTS AND SPECIFICATION

Project specification is a document used for effective project management that defines the overall plan, objectives, constraints, expected features, and system requirements. It helps ensure that all system functionalities are clearly identified and organized in a structured and accessible manner. The main objective of this phase is to collect and document all the requirements that the online voting system must fulfill.

This phase consists of two main activities:

- Requirement gathering and analysis
- Requirement specification

Requirement gathering and analysis

The goal of this stage is to collect relevant information from **election administrators and potential voters** regarding the online voting system to be developed. This helps ensure that system requirements are clearly understood, reducing errors, inconsistencies, and incomplete specifications.

This was achieved by:

- ❖ Interviewing election administrators
- ❖ Interviewing registered voters and potential users

Software requirement specification (SRS)

A **Software Requirement Specification (SRS)** forms the foundation of the entire project. It defines the framework that guides development, testing, quality assurance, and system maintenance. The SRS describes what the software will do, how it should perform, and the functionalities it must provide to meet stakeholder needs.

The requirements are divided into two main categories:

- ❖ Functional requirements.
- ❖ Non-functional requirements.

Functional requirements describe the technical operations and features of the online voting system. They define how users will interact with the system and what the system must be able to do. These may include:

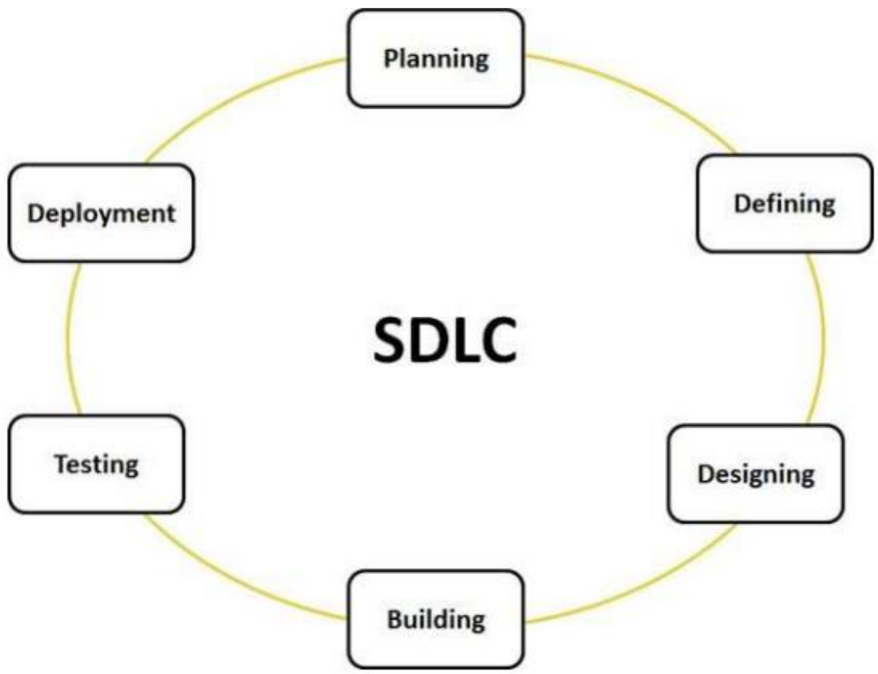
- ❖ User registration and authentication
- ❖ Vote casting and submission
- ❖ Vote storage and result computation
- ❖ Administrative management of voters and candidates

Non-Functional Requirements

Non-functional requirements specify **how the system should operate**, focusing on quality attributes rather than specific functions. They define system performance, security, usability, and reliability. These requirements describe how the system should behave rather than what it should do and are often referred to as **system attributes**.

SYSTEM DEVELOPMENT LIFE CYCLE (SDLC)

SDLC is a process followed for a software project, within a software organization. It consists of a detailed plan describing how to develop, maintain, replace and alter or enhance specific software. The life cycle defines a methodology for improving the quality of software and the overall development process. The following figure is a graphical representation of the various stages of a typical SDLC



AGILE MODEL:

The **Agile model** is a software development approach that focuses on **flexibility, continuous improvement, customer collaboration, and rapid delivery**. It breaks a project into **small, manageable iterations (sprints)**, allowing teams to develop, test, and improve the system gradually while adapting to changing user requirements.

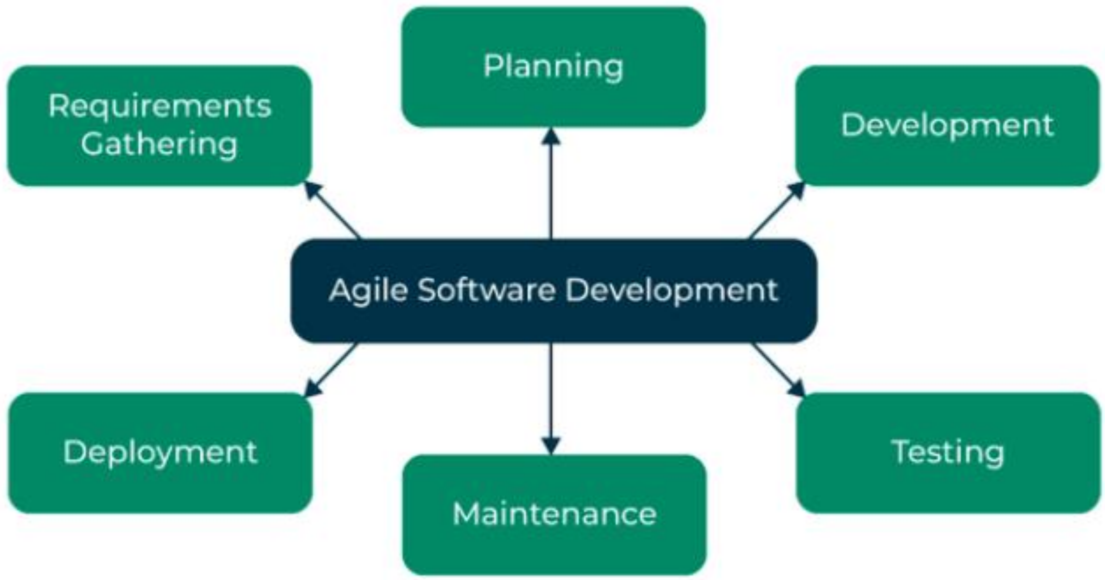
II.2.3. CHOICE OF MODEL

AGILE MODEL

The Agile model was chosen for this project because it allows flexibility, continuous feedback, and easy adaptation to changing requirements. It supports incremental development, enabling the system to be built, tested, and improved in small stages. This makes it suitable for the online voting system, where security, usability, and reliability must be refined throughout the development process.

WHY THE AGILE MODEL WAS SUITABLE FOR MY PROJECT

- **Incremental Development:** The system was developed in small parts (registration, login, voting, results). This made the project easier to manage and reduced complexity.
- **Flexibility to change:** Agile allows changes to be made at any stage. If new features or improvements were needed, they could be added without restarting the whole project.
- **Continuous Testing and Feedback:** Testing was done regularly during development. This helped identify errors early and improve the system based on feedback.
- **Early Error Detection:** Since testing happens after each iteration, problems were discovered and corrected quickly before becoming major issues.
- **Improved System Quality:** Continuous improvement and regular updates ensured the system became more reliable, secure, and user-friendly.
- **Better Project Control:** Working in short development cycles made it easier to track progress and manage tasks effectively.



IDENTIFICATION OF ACTORS

This project **"Online Voting System"** has two main actors: **Admin** and **Voter**, each with specific operations they can perform on the system.

Admin

- Can log in
- Can manage voter information
- Can manage candidates
- Can view election results

Voter

- Can register and log in
- Can cast a vote

- Can view election results

Candidates

- Can register and log in

Feasibility study

This study provided a complete overview of the project requirements to determine whether the **Online Voting System** is practical and achievable. The feasibility analysis focused on three major areas:

- **Economic feasibility:** Evaluates the cost of software development against the expected benefits of the system.
- **Technical feasibility:** Determines whether the required hardware, software, and technical skills are available and suitable.
- **Operational feasibility:** Assesses whether the system will operate effectively within its intended environment.

Review of feasibility study

Economic Feasibility

- Cost of software development
- Hosting plan
- Deployment cost

Technological Feasibility

- Programming skills
- System design skills

Operational Feasibility

- Voter registration
- Login and authentication
- Vote casting and result processing

Diagrams and definition

As earlier explained, an **Online Voting System** is an automated system that supports the management of voting activities for both administrators and voters. The system allows secure handling of voter registration, authentication, vote casting, and result processing.

The **admin** uses system features based on **CRUD operations** to manage system data. **CRUD** stands for **Create, Read, Update, and Delete**, which are the basic functions used to manage records such as voters, candidates, and election information within the system.

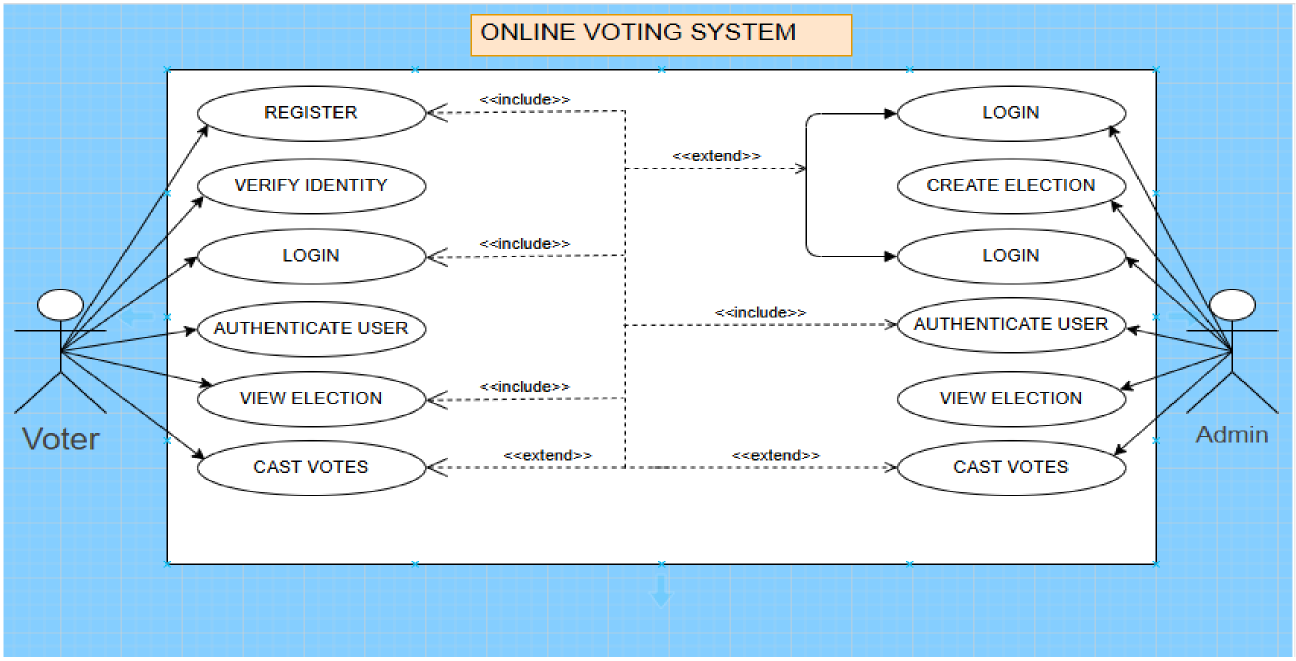


Figure 2: Use case diagram

In this use case, two people can interact with the system that is there are 2 actors in the system: The admin and the voter.

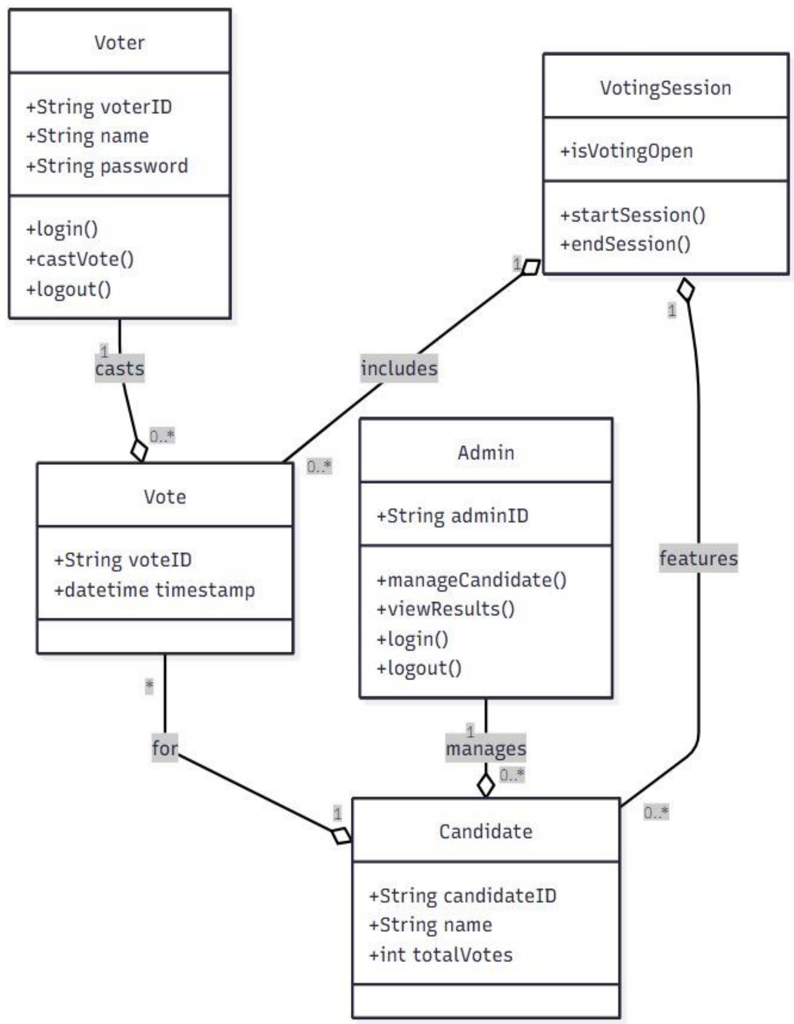


Figure 3: Class diagram

In the class diagram shown above, the **Voter** class has the attributes voterID, name, and password, and can perform operations such as login(), castVote(), and logout(), allowing the voter to participate in the election process. A voter can cast multiple votes, but each vote belongs to only one voter. The **Vote** class contains voteID and timestamp, representing each recorded vote, and each vote is associated with one candidate. The **Candidate** class has candidateID, name, and totalVotes, where totalVotes stores the number of votes received, and a candidate can receive many votes. The **Admin** class contains adminID and can perform operations such as manageCandidate(), viewResults(), login(), and logout(), enabling the admin to control candidates and monitor election results. The **VotingSession** class includes the attribute isVotingOpen and operations startSession() and endSession(), which control when voting is allowed. The voting session links the entire system by ensuring that votes are only cast when the session is active. Overall, the diagram shows the relationships and interactions between voters, votes, candidates, admin, and the voting session in the Online Voting System.

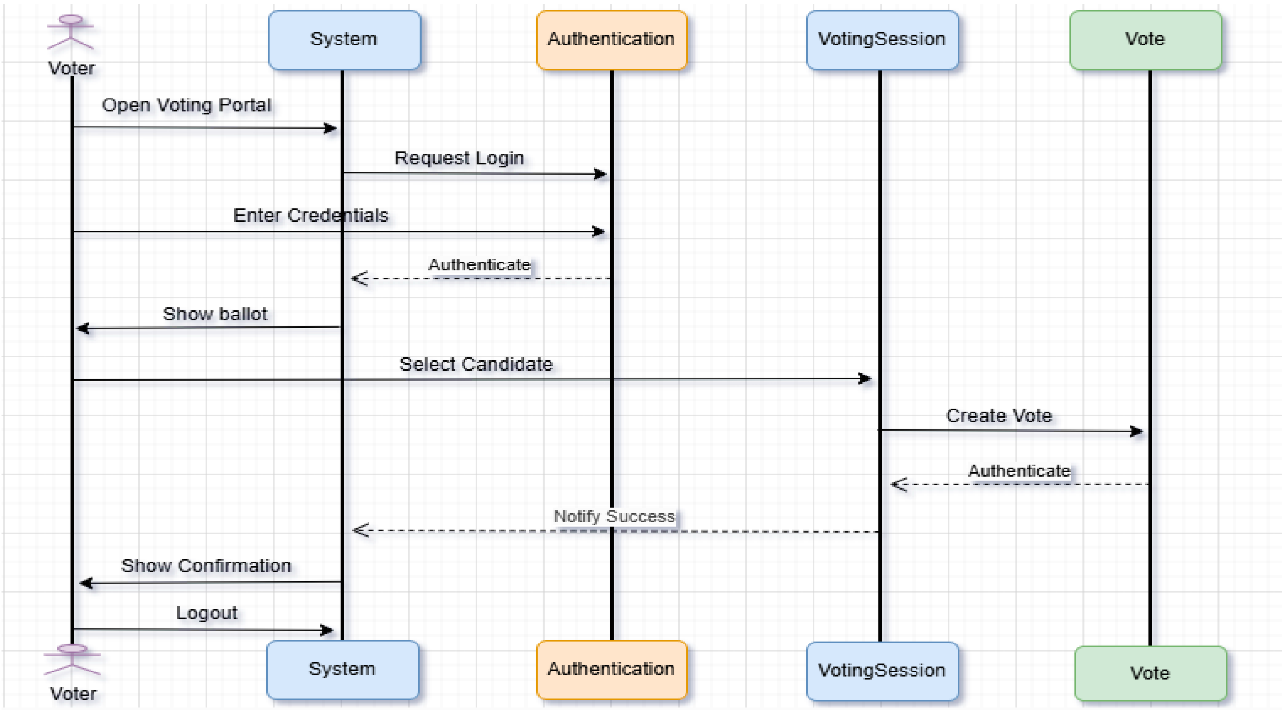


Figure 4: sequence diagram

The sequence diagram shows how the voter interacts with the system to cast a vote. The voter opens the voting portal, enters credentials, and the system sends them to the authentication module for verification before displaying the ballot. After selecting a candidate, the system creates the vote through the VotingSession and Vote components, then confirms successful submission. Finally, the system displays a confirmation message to the voter, who then logs out, completing the voting process.

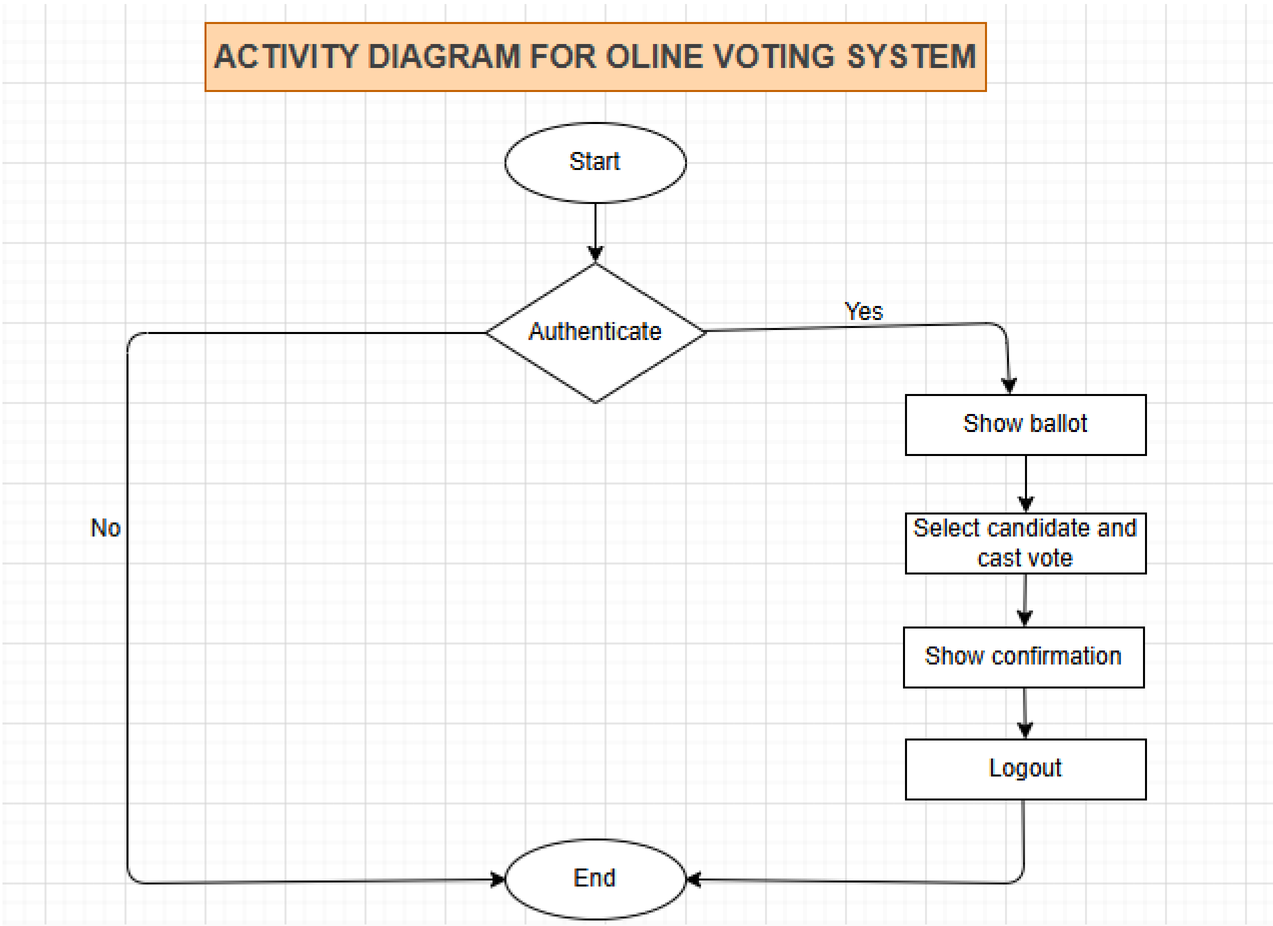


Figure 6: Activity diagram

Figure 7: Data flow diagram

Figure 9: ER diagram

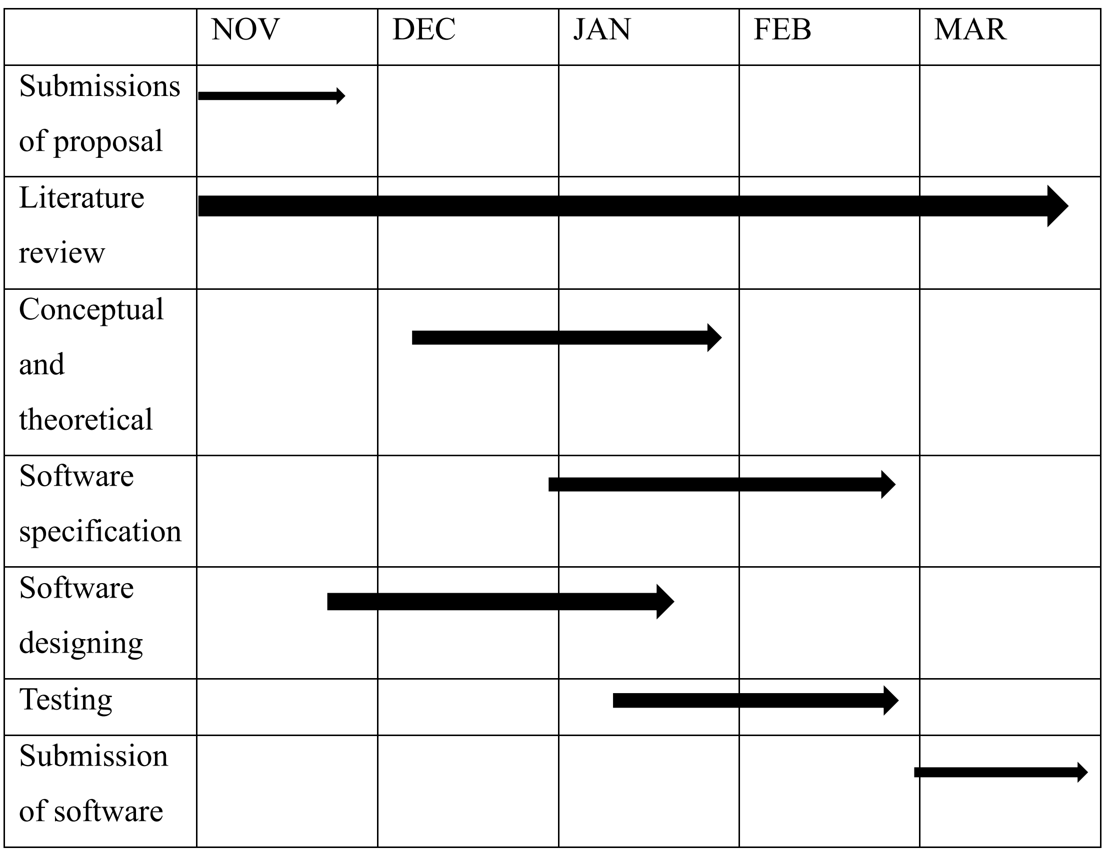


Figure 10: Gantt chart

Technologies used in the development of the project

Language: HTML, CSS, REACTJS.

Text editor: Visual studio code.

Database used: MongoDB.

Browser used: Chrome.

CHAPTER FOUR: RESULTS AND DISCUSSIONS

IV.1. PRESENTATION OF RESULT

INTERFACE OF THE SYSTEM FOR LOGIN FORM

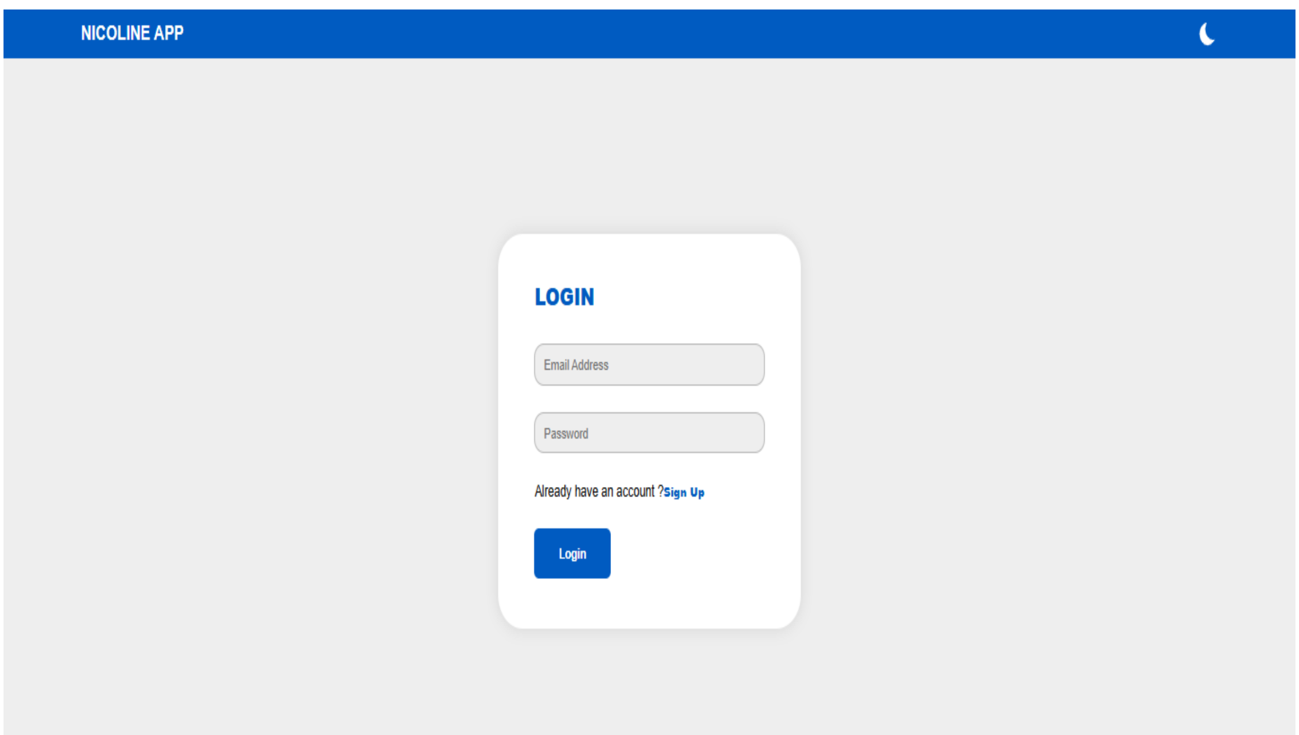


Figure 11: LOGIN FORM

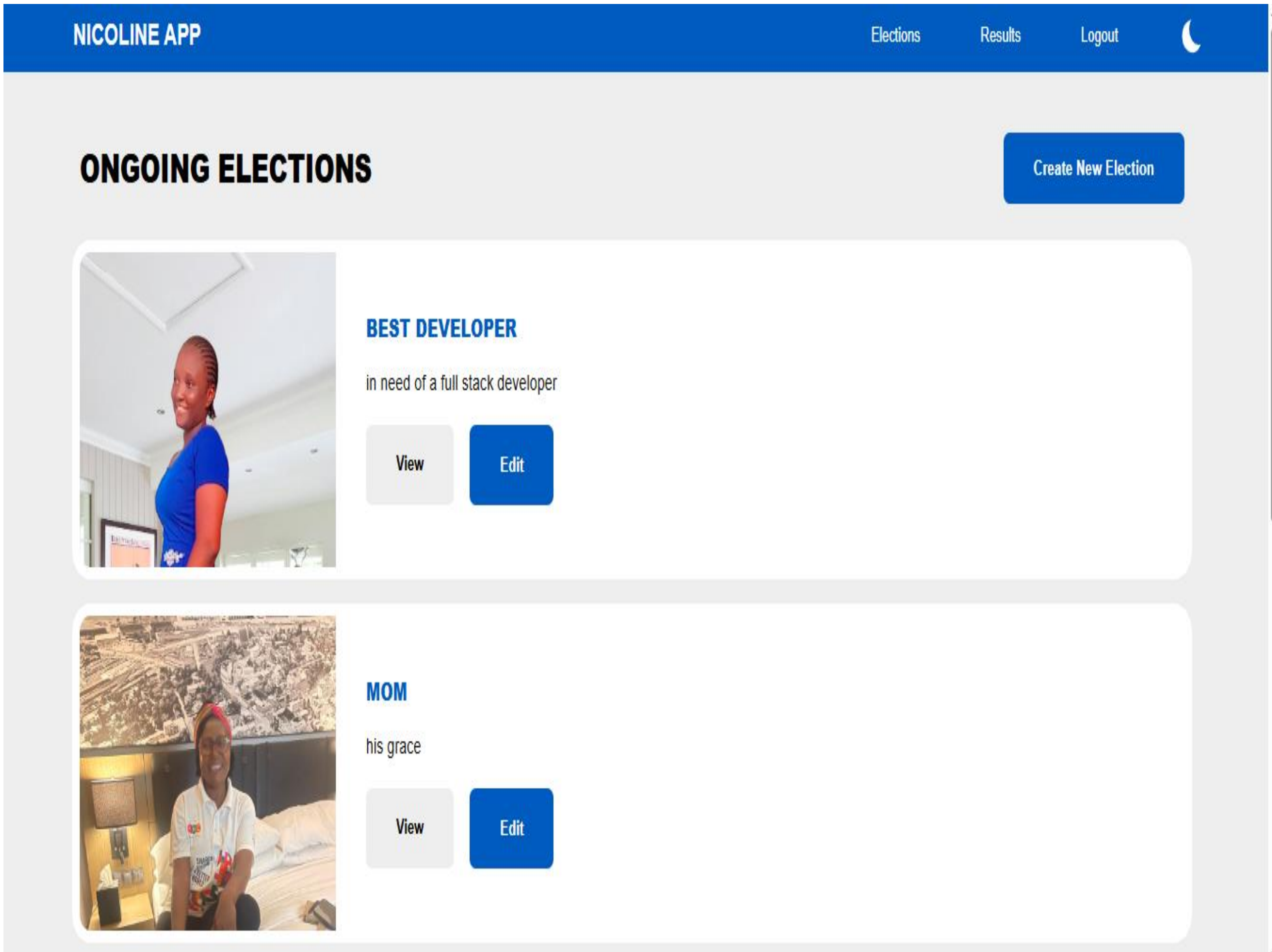


Figure 12: ADMIN DASHBOARD

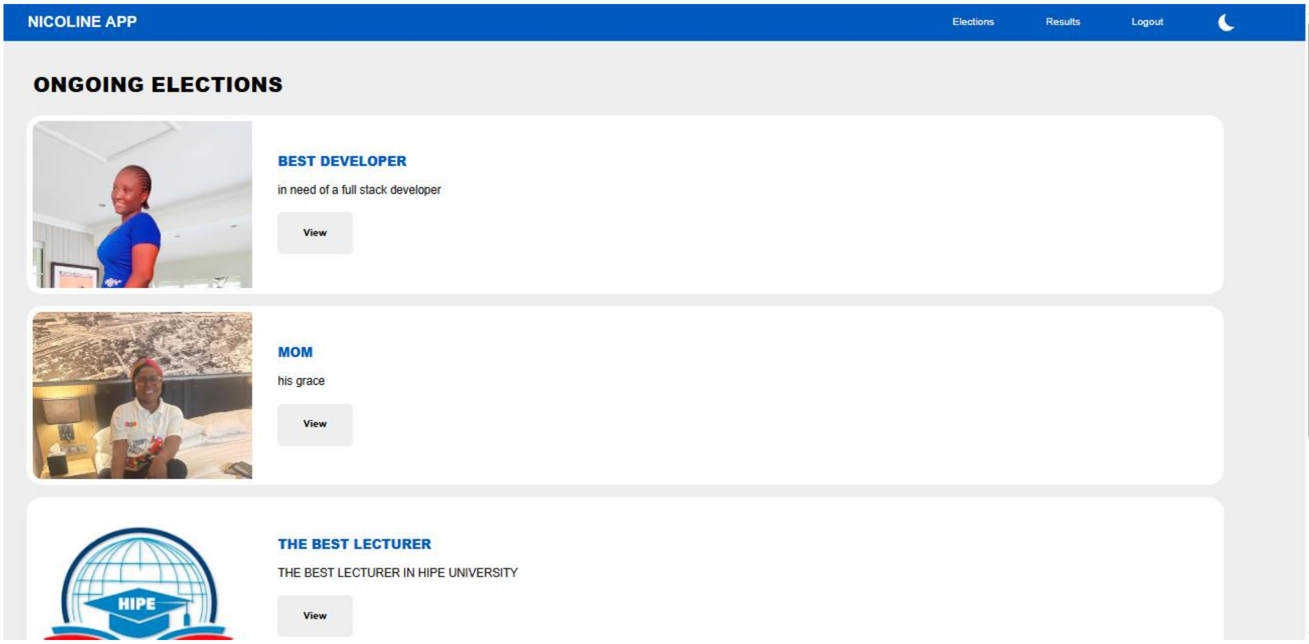


Figure 13: VOTER DASHBOARD

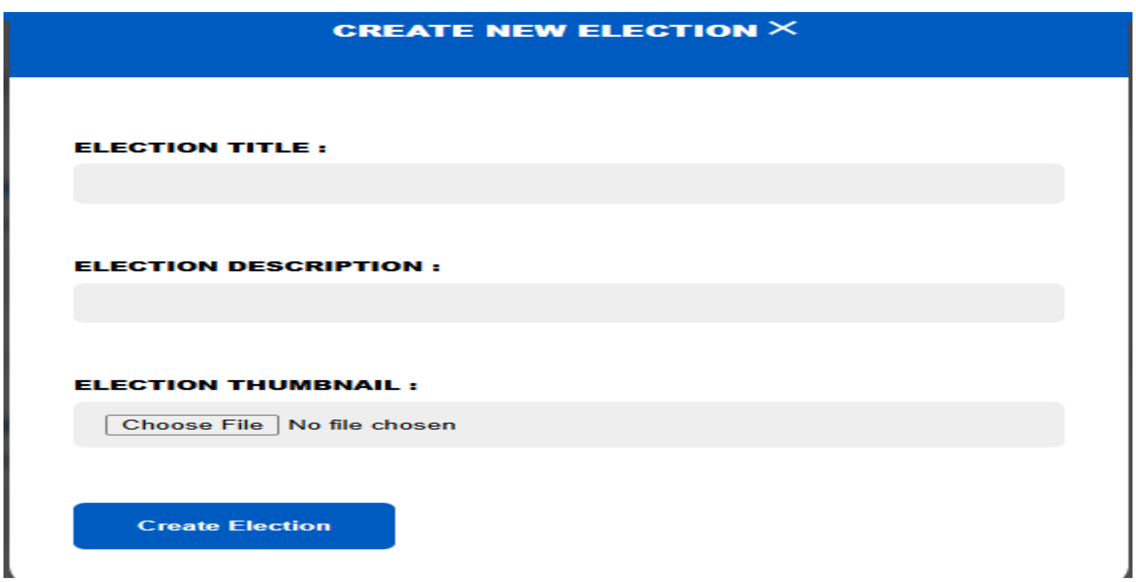


Figure 14: Create election page

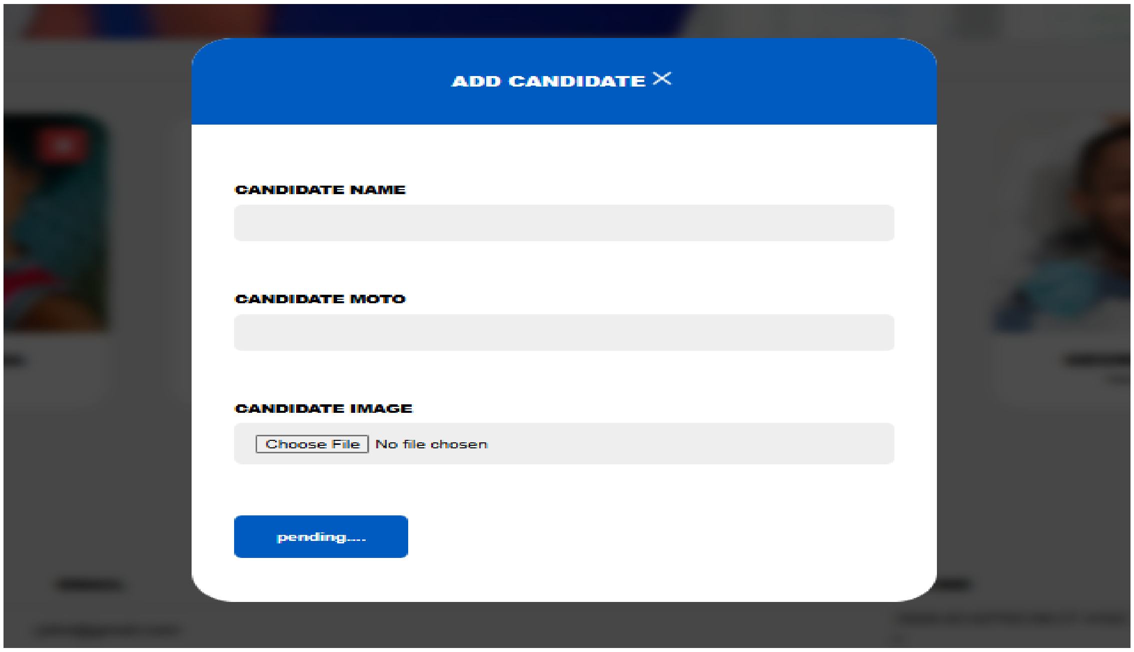


Figure 15: Add candidate page

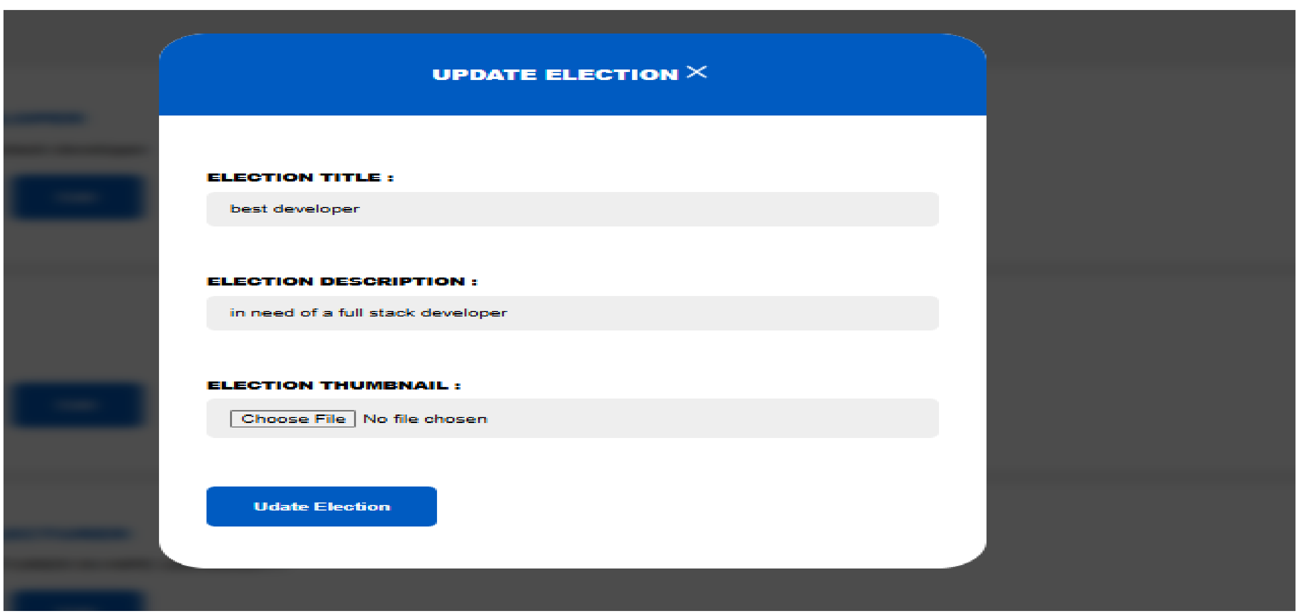


Figure 16: Update election page

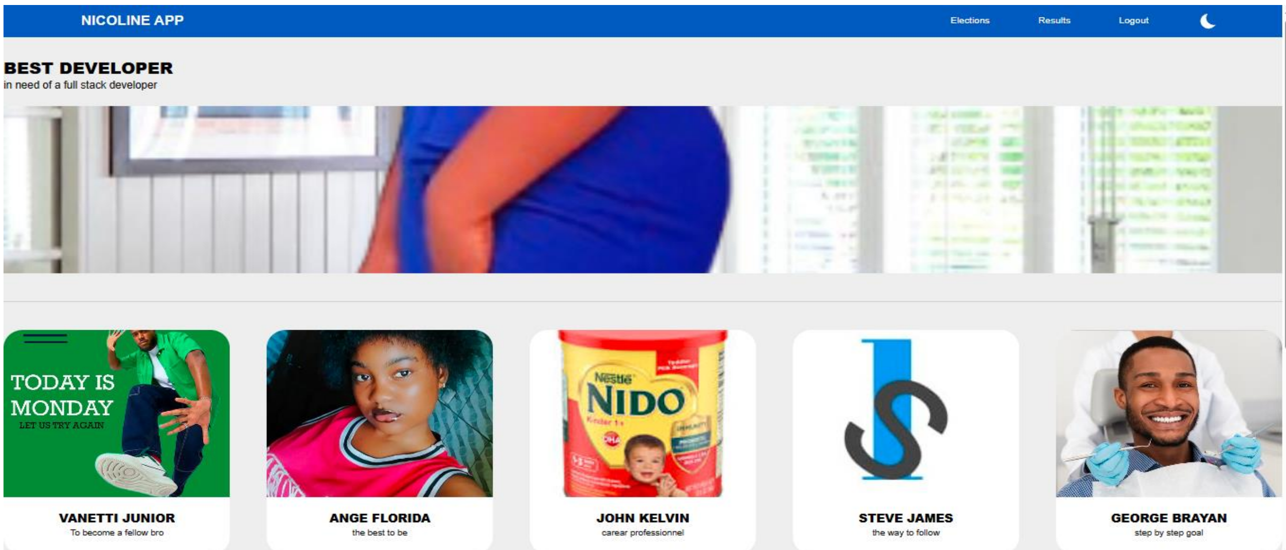


Figure 17: available candidate page for voters

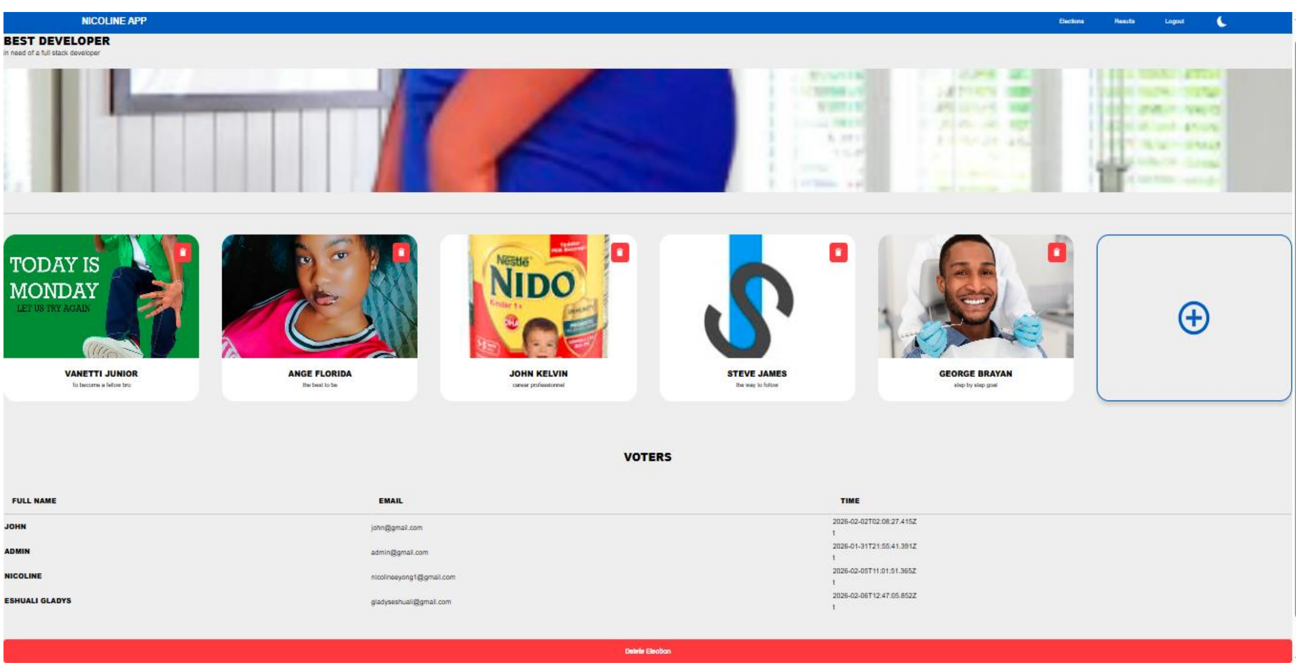


Figure 18: available candidates page for admin

IV.2. EXPLANATION OF RESULT

TESTING

Testing involves the execution of a software system or component to evaluate whether it meets specified requirements and is free from defects. It ensures that the developed Online Voting System performs correctly, securely, and efficiently. Testing may be carried out manually or using automated tools to evaluate different system properties. Below are the testing strategies used:

UNIT TESTING

Unit testing focuses on verifying the smallest components of the system such as login module, vote casting module, and result computation module. Each module is tested independently to ensure it functions correctly. This testing is performed during the coding phase.

VALIDATION TESTING

Validation testing ensures that the system meets user requirements. It confirms that the Online Voting System fulfills its intended purpose — allowing voters to cast votes securely and enabling the admin to manage the election process effectively.

PROVIDING TEST TOOLS

Test tools are software products that support testing activities such as planning, executing tests, logging defects, and analyzing results.

TOOL IMPLEMENTATION PROCESS

- Evaluating available testing tools and selecting suitable ones
- Analyzing strengths, weaknesses, and opportunities
- Developing a Proof of Concept
- Creating a pilot project
- Rolling out the tool gradually

The most preferable test tool for this project is a **Test Management Tool** because:

- It gathers and analyzes test data.
- Helps define requirements and manage test cases.
- Assists in planning test runs and analyzing results.
- Makes testing work easier and reusable.

GENERAL CONCLUSION

The development of the **Online Voting System** was aimed at facilitating secure, efficient, and transparent elections. The system is cost-effective, user-friendly, and reliable. Through this project, I improved my programming skills, database management knowledge, and user interface design abilities. Research and practical implementation greatly enhanced my understanding of software development processes.

The internship experience was highly beneficial, as it helped improve my technical and professional skills.

DIFFICULTIES ENCOUNTERED DURING THE INTERNSHIP

❖ **Technical issues**
Limited experience in programming languages such as PHP, Node.js, and JavaScript created challenges during development.

❖ **Social issues**
Working mostly with familiar colleagues limited exposure to diverse perspectives and reasoning styles.

❖ **Material issues**
Lack of a high-performance laptop affected productivity.

❖ **Financial issues**
Transportation costs and internet expenses posed financial challenges during the internship period.

IV.3. RECOMMENDATION AND PERSPECTIVES

Technology continues to evolve, and although the system meets its initial objectives, it can still be improved.

Future improvements may include:

- ❖ Enhancing system security with stronger encryption methods.
- ❖ Implementing multi-factor authentication.
- ❖ Improving system scalability and performance.

RECOMMENDATIONS

I recommend this Online Voting System to institutions, organizations, and schools that still conduct elections manually. The system reduces workload, improves transparency, saves time, and enhances election credibility.

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